

Distance-5 rotated surface code break-even crossing on

`ibm_kingston`

MWPM-decoded $d = 5$ Z-memory at $n_{\text{rounds}} = 1$ exceeds the wall-clock-matched bare-qubit baseline by $+2.88 \text{ pp} = 4.0\sigma$

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Abstract

We report a single-round logical-versus-physical break-even crossing for the distance-5 rotated surface code’s Z-memory on IBM Quantum’s `ibm_kingston` 156-qubit Heron-r2 processor. The experiment encodes $|0_L\rangle$ on a 5×5 data-qubit grid (25 data + 12 Z-stabilizer ancillas = 37 qubits), performs $n_{\text{rounds}} \in \{1, 2, 4\}$ rounds of Z-syndrome extraction, and decodes the syndrome difference history via minimum-weight perfect matching (MWPM) on a graph built with `networkx.min_weight_matching`. The MWPM-decoded logical recovery rate at $n_{\text{rounds}} = 1$ is $P(\hat{L}_{\text{MWPM}} = 0) = 0.8816$, compared with a wall-clock-matched bare-qubit baseline of $P_{\text{bare}}(Z = 0) = 0.8528$. The difference $+0.0288$ is 4.0σ above zero at 4096 shots. This is the first logical-error-rate below physical-error-rate demonstration in our Systrophē Heron-r2 QEC experimental program, and supersedes the $d = 3$ Steane sub-threshold baseline reported in `paper/steane_logical_qubit.pdf`. At higher n_{rounds} the MWPM advantage shrinks ($n=2$: -0.033 pp ; $n=4$: -0.020 pp) because our shortest-path decoder is approximate. We do *not* claim to beat the SOTA QEC threshold demonstrations of Google Quantum AI Willow (distance-7, sustained below-threshold) [?] or IBM bivariate-bicycle codes [?]; we report a clean single-round break-even crossing on Heron-r2 with a documented, reproducible MWPM pipeline.

1 Introduction

The defining hardware demonstration in quantum error correction is the *break-even* crossing: at matched wall-clock time, the encoded logical qubit retains Z (or X) information better than a bare physical qubit. Below this crossing the QEC overhead is net-harmful; above it, encoding pays off. The classical milestone is Google Quantum AI’s distance-7 Willow result (Dec 2024) [?] showing sustained below-threshold scaling. Earlier IBM and Google distance-3 to distance-5 comparisons found a fragile crossing only at distance-5, with very specific decoder choices and round counts [?].

In our companion paper [?] we reported a $d = 3$ Steane `[[7,1,3]]` round-sweep experiment on `ibm_kingston` with $T_{1,L}/T_{1,\text{phys}} = 0.579$ — a clean sub-threshold result. This paper reports the natural follow-up: a $d = 5$ rotated surface code on the same chip with the same wall-clock-matched baseline methodology. We find a single-round break-even crossing at 4.0σ , but no crossing at higher round counts within our naive MWPM implementation.

2 Experiment

2.1 Code and circuit

We use the canonical rotated $d = 5$ surface code with rough top/bottom boundaries (X-edges) and smooth left/right boundaries (Z-edges) on a 5×5 data-qubit grid (25 data qubits). The Z-stabilizers comprise eight weight-4 plaquettes plus four weight-2 boundary half-plaquettes, totalling $N_Z = 12$. We additionally allocate 12 Z-ancilla qubits for syndrome extraction (one per Z-stab), giving a total qubit count of 37.

For the Z-memory experiment, encoding $|0_L\rangle$ is trivial: the data register initialised to $|0\rangle^{25}$ is automatically the $+1$ eigenstate of all Z-stabilizers and has $Z_L = +1$. No explicit encoder circuit is required when only Z-syndromes are extracted (which suffices for protection against X-type errors). Each round of Z-syndrome extraction applies, for each Z-stab, CNOTs from each data qubit in the stab’s support into the dedicated ancilla, followed by ancilla measurement and reset.

We do not extract X-syndromes in this experiment, so X-type errors propagating into Z observables during measurement are not corrected. This is a deliberate trade-off: the Z-memory experiment is the simplest valid $d = 5$ surface-code memory test, and the X-syndrome extension is the natural next step.

2.2 Submission

The full round sweep was submitted as a single SamplerV2 batch (job `d81ke16gbeec73akmp00`, 4096 shots/circuit) on `ibm_kingston` with:

- Optimization level 3, ISA-transpiled to Heron-r2 heavy-hex topology.
- Dynamical decoupling enabled (XpXm sequence).
- Three Z-memory circuits at $n_{\text{rounds}} \in \{1, 2, 4\}$, paired with three bare-qubit baselines at matched compiled durations Δt (using `qc.delay(dt)` on a single qubit + measurement).

ISA depths and durations:

n_{rounds}	Surface ISA depth	Duration (dt units)	Bare ISA depth
1	152	30 400	2
2	289	57 800	2
4	465	93 000	2

Note that the $d = 5$ surface code’s ISA depth at $n_{\text{rounds}} = 1$ (152) is *shallower* than our $d = 3$ Steane code at the same round count (229), because the surface code’s Z-syndrome circuit is parallel-friendly (no H -wrapping required) and the heavy-hex transpiler routes the surface code’s local nearest-neighbour CNOTs more efficiently than the Steane code’s non-local stabilizer support.

2.3 Minimum-weight perfect matching decoder

The decoder takes the syndrome difference history across rounds plus the final data-derived Z-syndrome and decodes via `networkx.min_weight_matching` on a graph with one node per violated stabilizer per round. Edges are weighted by lattice-Manhattan distance between stab centers. For each matched pair, we apply X -flip corrections to the shortest-path data qubit (or the shared-support qubit if any). Unpaired (odd-cardinality) violations are mapped to their first-support qubit.

This is an approximate MWPM: a fully fault-tolerant implementation would use a 3D syndrome graph (round $\times$ stab $\times$ stab) with Dijkstra-shortest-path edge weights and explicit boundary-edge nodes. Our simplified implementation suffices to demonstrate the single-round break-even crossing reported here.

We compare the MWPM decoder to a baseline “greedy” decoder that uses only the final data-derived Z-syndrome and applies a fixed correction per violated stab. The greedy decoder is essentially the same as the $d = 3$ Steane lookup-table decoder; for $d = 5$ it cannot resolve multi-qubit error chains and performs worse than MWPM.

3 Results

n_{rounds}	Greedy	MWPM	Bare baseline	MWPM – bare	Significance
1	0.6990	0.8816	0.8528	+0.0288	4.0σ
2	0.6013	0.7429	0.7754	−0.0325	4.0 σ
4	0.5337	0.6821	0.7024	−0.0203	2.4 σ

The headline numbers:

- **At $n_{\text{rounds}} = 1$, the MWPM-decoded $d = 5$ surface code Z-memory rate is 0.8816, exceeding the wall-clock-matched bare-qubit baseline of 0.8528 by +2.88 percentage points (4.0 σ at 4096 shots).**
- The greedy decoder, by comparison, gives only 0.699 at $n_{\text{rounds}} = 1$ — 15.4 pp below bare. MWPM is critical for recovering the $d = 5$ advantage.
- At $n_{\text{rounds}} \geq 2$, MWPM still loses to bare, with the gap shrinking as n grows (from −3.25 pp at $n = 2$ to −2.03 pp at $n = 4$). The MWPM advantage at single-round measurement does not yet sustain across multiple rounds.

3.1 Why the advantage does not extend past $n = 1$

Two reasons:

1. The shortest-path X-flip correction we apply between matched stab pairs is a one-qubit flip, not the full multi-qubit chain along the lattice-Manhattan shortest path. For weight-2 distances this is correct; for larger distances it under-corrects, growing in error rate as the round count and per-round error rate grows.
2. Z-syndrome-only memory does not detect phase errors that flip the X-eigenstate of the measurement ancillas during their measure-reset cycle. These propagate forward and accumulate.

A full fault-tolerant MWPM decoder with proper 3D syndrome graphs and boundary nodes would likely extend the advantage further. We defer this to a follow-up study.

3.2 Comparison to literature SOTA

Google Quantum AI’s distance-3 vs distance-5 surface code comparison in 2023 [?] reported a similar single-round-style crossing that disappeared at higher rounds without further architectural improvements. The Willow result [?] achieved sustained sub-threshold scaling at distance-7 with much higher gate fidelities (cz error $\sim 5 \times 10^{-4}$ vs our Heron-r2 $\sim 2 \times 10^{-3}$). Our result is consistent with the literature picture: $d = 5$ on current Heron-r2 gate fidelities lives near the threshold; with

proper MWPM at single-round measurement we cross break-even; sustained crossing across rounds requires either gate-fidelity improvement (Willow-class) or larger distance ($d \geq 7$).

Honest claim-level (Systrophē taxonomy):

- Reproducible: yes; rerun `python experiments/surface_code_d5_mwpm.py` or resubmit via `surface_code_d5_round_sweep.py --submit --n-rounds 1`.
- Novel demonstration on Heron-r2: yes; first single-round $d = 5$ surface-code break-even crossing in our experimental program.
- SOTA contribution: no; this does not improve on Willow or BB-code published thresholds.

4 Open extensions

1. **Full fault-tolerant MWPM** with 3D syndrome graphs and boundary nodes (Dennis-Kitaev-Landahl-Preskill 2002).
2. **X + Z syndrome extraction** (full surface code memory, not just Z).
3. **Higher distance** ($d = 7$ would need 49 data + 24 ancillas = ~ 90 qubits, still fits Heron-r2’s 156).
4. **More shots** at $n_{\text{rounds}} = 1$ to push the break-even-crossing significance from 4.0σ to $10\sigma+$.
5. **Multi-logical-qubit demonstration:** parallel patches of the same code on disjoint Heron-r2 subgraphs.

5 Reproducibility

All code and data are at github.com/Zynerji/systrophe:

- Circuit construction: `experiments/surface_code_d5.py`
- Round sweep submission: `experiments/surface_code_d5_round_sweep.py`
- MWPM decoder: `experiments/surface_code_d5_mwpm.py`
- Job ID: `d81kel16gbeec73akmp00` (round sweep + bare baselines)
- Raw analysis JSON: `experiments/results/surface_code_d5_mwpm_analysis.json`

References

- [1] Google Quantum AI. *Quantum error correction below the surface code threshold*. Nature **636**, 456 (Dec 2024).
- [2] Google Quantum AI. *Suppressing quantum errors by scaling a surface code logical qubit*. Nature **614**, 676 (2023).
- [3] S. Bravyi et al. *High-threshold and low-overhead fault-tolerant quantum memory*. Nature **627**, 778 (2024).
- [4] C. Knopp. *Distance-3 Steane code repeated-round logical-qubit experiment on ibm_kingston*. Systrophē repository, `paper/steane_logical_qubit.pdf` (2026).